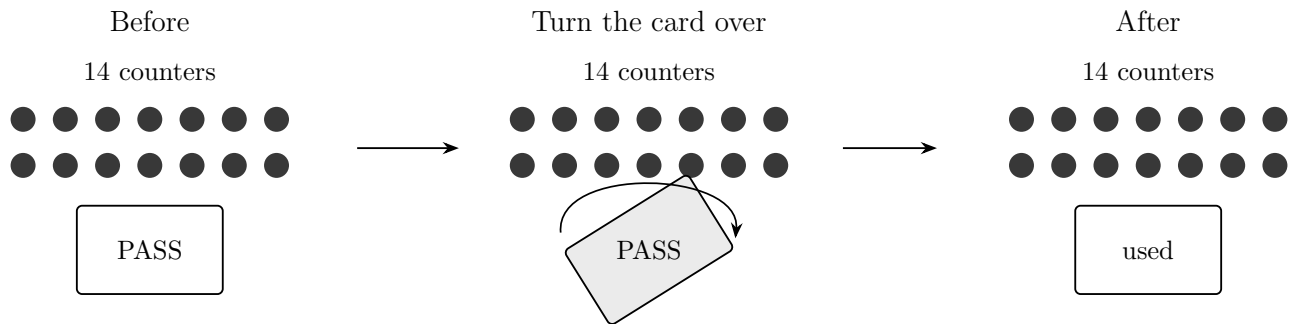


Take turns taking 1 or 2 counters. Either player may turn the shared face-up PASS card face down instead of taking counters. Taking the last counter ends the game and wins.

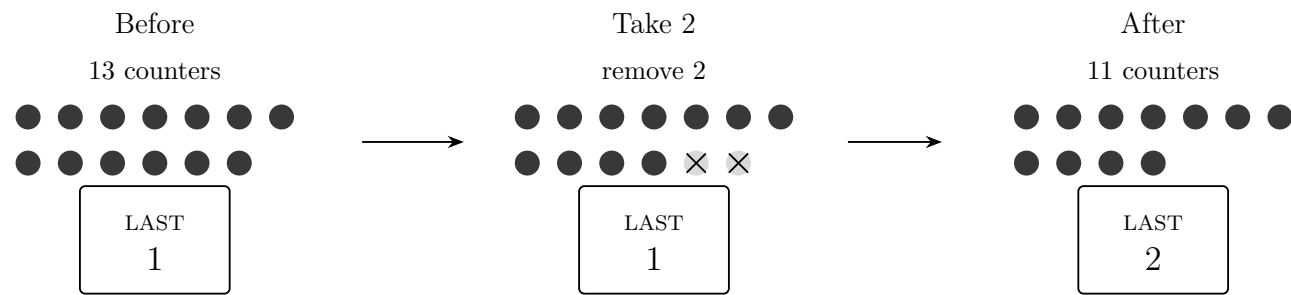


Problem 1: Play with 1 to 12 counters. When does one PASS card change which player can make sure they win? Circle the starts where the first player can make sure they win in each list.

PASS face up			
1	2	3	4
5	6	7	8
9	10	11	12

PASS already used			
1	2	3	4
5	6	7	8
9	10	11	12

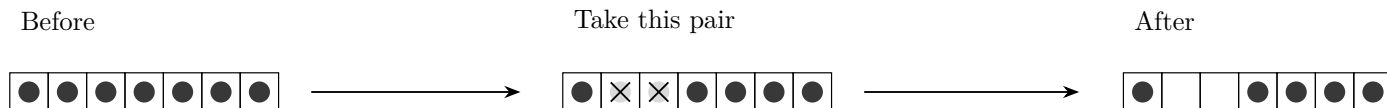
Take turns taking 1, 2 or 3 counters. You may not take the number on the face-up LAST card. After each turn, turn the old LAST card face down and the matching card face up. Taking the last counter wins; if you cannot make a legal move, you lose. No face-up LAST card means there is no previous move.



Problem 2: Can changing the LAST card change whether the first player can make sure they win? Play with piles of 1 to 8 counters and circle the winning starts in each column. Find a rule that predicts the winners for larger piles.

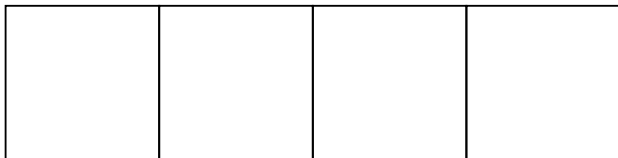
No previous move	LAST 1	LAST 2	LAST 3
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Take turns removing exactly 2 counters from neighbouring squares in one row. Gaps stay empty; counters do not slide. If you cannot take a pair, you lose.

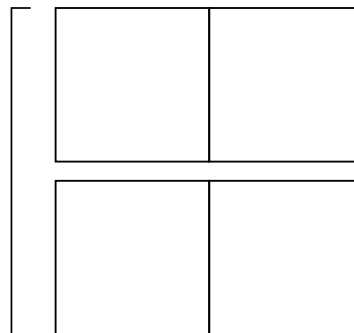


Problem 3: Choose a start and put a counter in each square. Play all six starts and circle those where the first player can make sure they win.

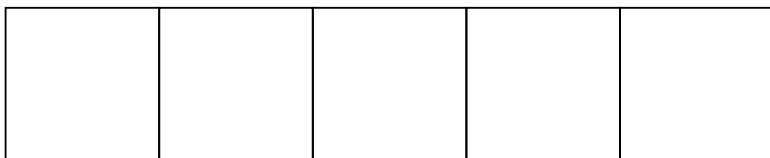
A



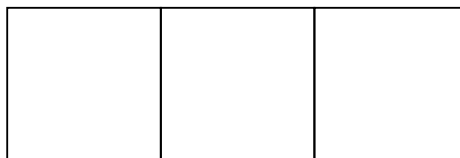
B



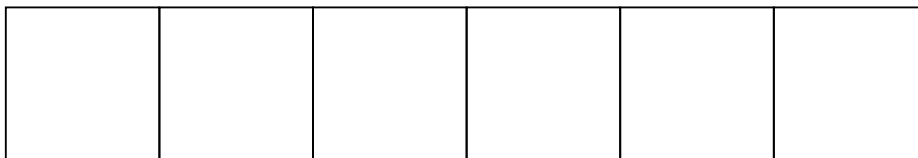
C



D



E



F

